

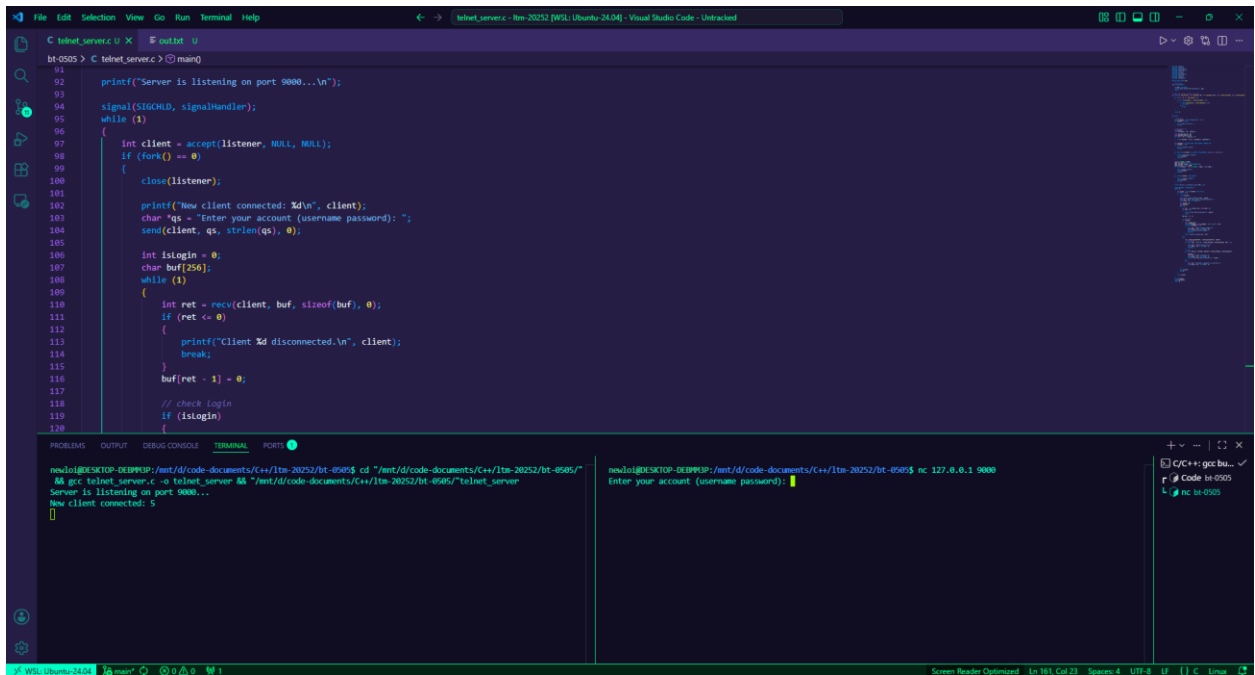
BÀI TẬP VỀ NHÀ

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Bài 1: telnet_server

- Chạy telnet_server, dùng netcat client để kết nối server



The screenshot displays the Visual Studio Code editor with the `telnet_server.c` file open. The code is a C program that listens on port 9000 for incoming connections. It uses `accept()` to handle connections, prints the client IP, and prompts the user to enter an account name and password. It also includes a loop to receive data from the client and a check for login status.

```
91 // Server is listening on port 9000...
92 printf("Server is listening on port 9000...\n");
93
94 signal(SIGCHLD, signalHandler);
95 while (1)
96 {
97     int client = accept(listener, NULL, NULL);
98     if (fork() == 0)
99     {
100         close(listener);
101         printf("New client connected: %d\n", client);
102         char qs = "Enter your account (username password): ";
103         send(client, qs, strlen(qs), 0);
104
105         int islogin = 0;
106         char buf[256];
107         while (1)
108         {
109             int ret = recv(client, buf, sizeof(buf), 0);
110             if (ret <= 0)
111             {
112                 printf("Client %d disconnected.\n", client);
113                 break;
114             }
115             buf[ret - 1] = 0;
116
117             // check login
118             if (islogin)
119             {
120                 // ...
121             }
122         }
123     }
124 }
```

The terminal window shows the execution of the program. It starts by printing "Server is listening on port 9000...". Then, it receives a connection from a netcat client (nc 127.0.0.1 9000). The server prompts the user to enter an account name and password. The client enters "5", and the server prints "New client connected: 5".

- Client đăng nhập vào server (trường hợp thông tin tài khoản sai, hoặc gửi lệnh trước khi đăng nhập)

The screenshot shows the Visual Studio Code editor with a C++ file named `accounts.txt` open. The file contains a list of usernames and passwords:

```
1 3
2 admin admin
3 newadmin 123456789
4 guest 123456
```

Below the editor, the **TERMINAL** panel is active, displaying the output of a C++ program. The program is a telnet server that listens on port 9000 and handles client connections. The output shows the server starting, listening, and receiving a connection from a client with IP 127.0.0.1. The client then enters the username 'admin' and the password 'admin', which are compared against the data in `accounts.txt`. The output shows that the password is incorrect.

```
newid@DESKTOP-DEIPEPSP:/mnt/d/code-documents/C++/Itm-20252/bt-0905$ cd "/mnt/d/code-documents/C++/Itm-20252/bt-0905/"
$ g++ telnet_server.cpp o telnet_server $ gcc telnet_server
Server is listening on port 9000...
New client connected: 5
admin
admin
Please login first.
admin admin
Please login first.
admin ad
Username or password is incorrect.
```

The status bar at the bottom indicates the file is `accounts.txt` in the `Itm-20252` directory, with the language set to C++.

The image shows a Visual Studio Code editor window with a C++ project named 'accounts.txt'. The file 'accounts.txt' contains the following content:

```
1 3
2 admin admin
3 newdoi loungecloi
4 guest loungecloi
```

The terminal output shows the execution of the program. The user runs the command `nc 127.0.0.1 9905` to connect to the telnet server. The server output is as follows:

```
newdoi@DESKTOP-DEIWPSP:/mnt/d/code-documents/C++/ltn-20252/bt-6905$ cd "/mnt/d/code-documents/C++/ltn-20252/bt-6905/"
$ g++ telnet_server.cpp o telnet_server $ gcc telnet_server $ gcc telnet_server
Server is listening on port 9900...
New client connected: 5
Client 5 login successful.
```

The terminal also shows the output of the program when it receives a connection from the netcat client:

```
newdoi@DESKTOP-DEIWPSP:/mnt/d/code-documents/C++/ltn-20252/bt-6905$ nc 127.0.0.1 9905
Enter your account (username password): admin
Please login first.
admin admin admin
Please login first.
admin ad
Username or password is incorrect.
admin admin
Login successful.
```

```
bt-0505 > F out.txt
1
2
3 accounts.txt
4 out.txt
5 telnet_server
6 telnet_server.c
7

newdoi@DESKTOP-0E8PWP3P:/mnt/d/code-documents/C++/ltn-20252/bt-0505$ nc 127.0.0.1 9000
Enter your account (username password): admin
Please login first.
admin admin admin
Please login first.
admin ad
Username or password is incorrect.
admin admin
Login successful.
ls -a
```

Bài 2: http_server

- Chạy http_server và 2 client netcat kết nối tới server sẽ nhận được response
 - o Client 1

```
53     exit(1);
54 }
55
56 printf("Server is listening on port 9000...\n");
57
58 // handler close process
59 signal(SIGCHLD, signalhandler);
60 while (1)
61 {
62     int client = accept(listener, NULL, NULL);
63     if (fork() == 0) // on child process
64     {
65         close(listener);
66
67         printf("New request from client: %d\n", client);
68
69         // return response to client
70         char *msg = "HTTP/1.1 200 OK\r\nContent-type: text/html\r\nContent-Length: 34\r\n\r\n<html><body><h1>Hello World</h1></body></html>";
71         send(client, msg, strlen(msg), 0);
72         close(client);
73         exit(0);
74     }
75     close(client);
76 }
77
78 close(listener);
79
80 return 0;
```

```
newdoi@DESKTOP-0E8PWP3P:/mnt/d/code-documents/C++/ltn-20252$ nc 127.0.0.1 9000
HTTP/1.1 200 OK
Content-Type: text/html
<html><body><h1>Hello World</h1></body></html>
```

- o Client 2

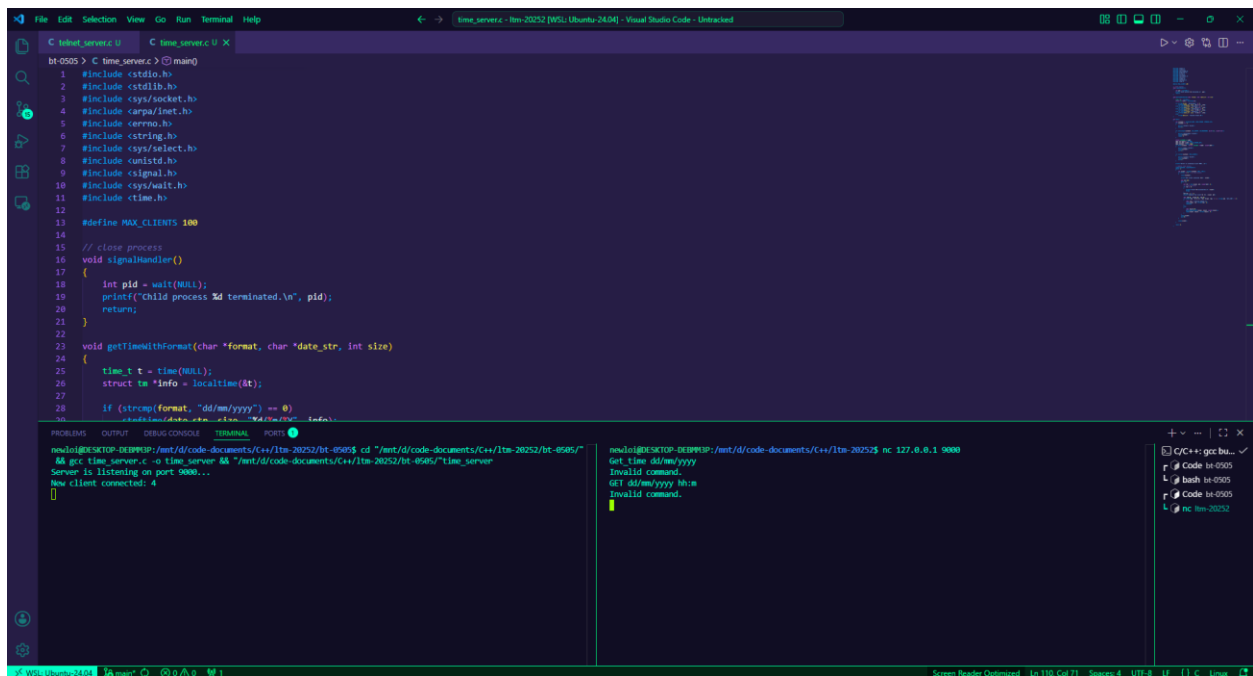
The screenshot shows the Visual Studio Code editor with the file `http_server.c` open. The code is a C program that listens on port 9000 and responds to HTTP requests with a "Hello World" message. The output window shows the server's execution, including the message "Server is listening on port 9000..." and the received request from client 4. The terminal window shows the command `nc 127.0.0.1 9000` being executed, and the output shows the received request and the response "Hello World".

Bài 3: time_server

- Chạy time_server và client netcat

The screenshot shows the Visual Studio Code editor with the file `time_server.c` open. The code is a C program that listens on port 9000 and responds to requests with the current time. The output window shows the server's execution, including the message "Server is listening on port 9000..." and the received request from client 4. The terminal window shows the command `nc 127.0.0.1 9000` being executed, and the output shows the received request and the response "The time is: 10:10:10".

- Client gửi lệnh tới server
 - o Sai lệnh

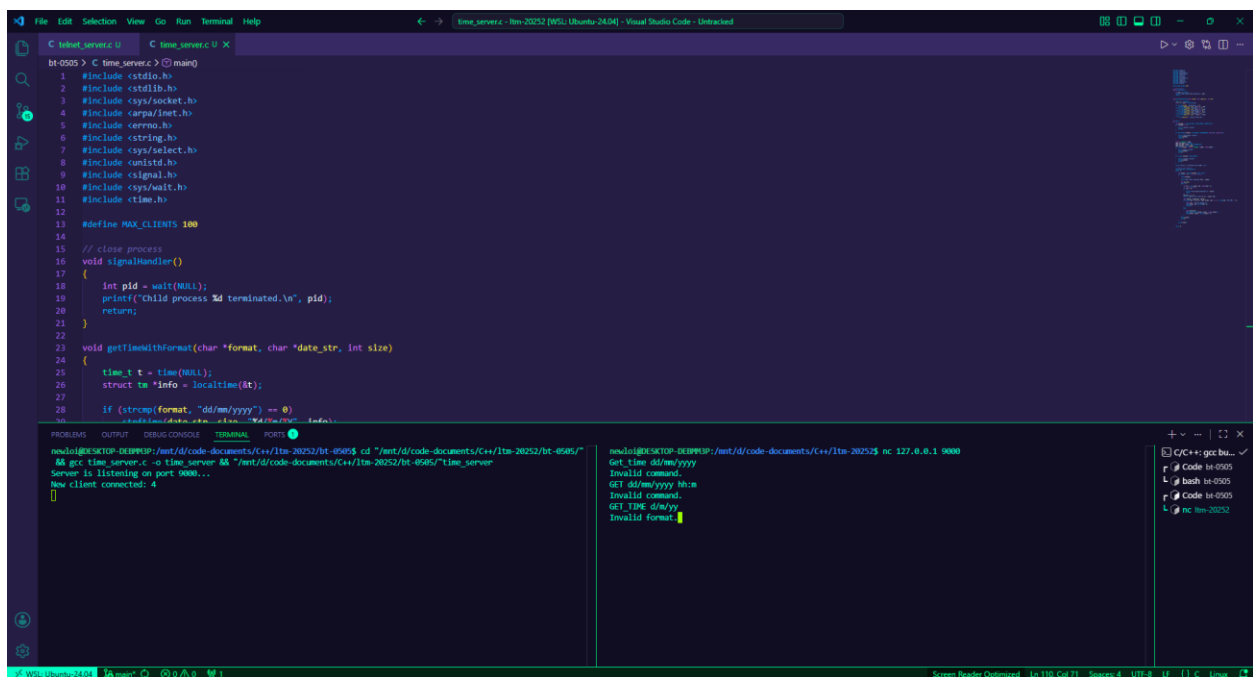


```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <sys/socket.h>
4 #include <arpa/inet.h>
5 #include <errno.h>
6 #include <string.h>
7 #include <sys/select.h>
8 #include <unistd.h>
9 #include <signal.h>
10 #include <sys/wait.h>
11 #include <time.h>
12
13 #define MAX_CLIENTS 100
14
15 // close process
16 void signalHandler()
17 {
18     int pid = wait(NULL);
19     printf("Child process %d terminated.\n", pid);
20     return;
21 }
22
23 void gettimeofdayFormat(char *format, char *date_str, int size)
24 {
25     time_t t = time(NULL);
26     struct tm *info = localtime(&t);
27
28     if (strftime(format, "dd/mm/yyyy" == 0)
29         strftime(date_str, size, "%d/%m/%Y", info);
30 }
```

```
newui@DESKTOP-DEBPPUP:/mnt/d/code-documents/C++/itm-20252/bt-4905$ cd "/mnt/d/code-documents/C++/itm-20252/bt-4905/"
$ gcc time_server.c -o time_server $& "/mnt/d/code-documents/C++/itm-20252/bt-4905/"time_server
Server is listening on port 9000...
New client connected: 4
```

```
newui@DESKTOP-DEBPPUP:/mnt/d/code-documents/C++/itm-20252$ nc 127.0.0.1 9000
Get Time dd/mm/yyyy
Invalid command.
GET dd/mm/yyyy hha
Invalid command.
GET TIME d/m/y
Invalid format.
```

- Sai định dạng thời gian



```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <sys/socket.h>
4 #include <arpa/inet.h>
5 #include <errno.h>
6 #include <string.h>
7 #include <sys/select.h>
8 #include <unistd.h>
9 #include <signal.h>
10 #include <sys/wait.h>
11 #include <time.h>
12
13 #define MAX_CLIENTS 100
14
15 // close process
16 void signalHandler()
17 {
18     int pid = wait(NULL);
19     printf("Child process %d terminated.\n", pid);
20     return;
21 }
22
23 void gettimeofdayFormat(char *format, char *date_str, int size)
24 {
25     time_t t = time(NULL);
26     struct tm *info = localtime(&t);
27
28     if (strftime(format, "dd/mm/yyyy" == 0)
29         strftime(date_str, size, "%d/%m/%Y", info);
30 }
```

```
newui@DESKTOP-DEBPPUP:/mnt/d/code-documents/C++/itm-20252/bt-4905$ cd "/mnt/d/code-documents/C++/itm-20252/bt-4905/"
$ gcc time_server.c -o time_server $& "/mnt/d/code-documents/C++/itm-20252/bt-4905/"time_server
Server is listening on port 9000...
New client connected: 4
```

```
newui@DESKTOP-DEBPPUP:/mnt/d/code-documents/C++/itm-20252$ nc 127.0.0.1 9000
Get Time dd/mm/yyyy
Invalid command.
GET dd/mm/yyyy hha
Invalid command.
GET TIME d/m/y
Invalid format.
```

- Đúng lệnh và định dạng

The screenshot displays a C++ development environment with a source code editor at the top and a multi-paneled output window at the bottom.

Source Code Editor:

```
#include <iostream>
#include <string>
#include <unistd.h>
#include <sys/types.h>
#include <time.h>
#include <signal.h>

using namespace std;

int main()
{
    // close process
    void signalHandler()
    {
        int pid = wait(NULL);
        printf("Child process %d terminated.\n", pid);
        return;
    }

    void getLineWithFormat(char *format, char *date_str, int size)
    {
        time_t t = time(NULL);
        struct tm *info = localtime(&t);

        if (strcmp(format, "dd/mm/yyyy") == 0)
            strftime(date_str, size, "%d/%m/%Y", info);
        else if (strcmp(format, "dd/mm/yy") == 0)
            strftime(date_str, size, "%d/%m/%y", info);
        else if (strcmp(format, "mm/dd/yyyy") == 0)
            strftime(date_str, size, "%m/%d/%Y", info);
        else if (strcmp(format, "%m/%d/%y") == 0)
            strftime(date_str, size, "%m/%d/%y", info);
        else
            strcpy(date_str, "Invalid format.%B");
    }
}
```

Output Window:

The output window shows the execution results across four terminal panes:

- Terminal 1:** Shows the compilation command:
`g++ ./tme-20252/bt-49495.c -o "/mnt/d/code-documents/C++/Itm-20252/bt-49495/" && gcc time\_server.c -o time\_server /usr/include/time-server`
And the output:
Server is listening on port 9000...
New client connected: 4
New client connected: 4
New client connected: 4
New client connected: 4
- Terminal 2:** Shows the execution of the compiled binary:
`./tme-20252/bt-49495`
Output:
GET_TIME dd/mm/yy
18/05/2024
- Terminal 3:** Shows the execution of the compiled binary:
`./tme-20252/bt-49495`
Output:
GET_TIME mm/dd/yyyy
05/18/2024
- Terminal 4:** Shows the execution of the compiled binary:
`./tme-20252/bt-49495`
Output:
GET_TIME dd/mm/yy
18/05/24

The status bar at the bottom indicates the file path as `C:\Users\user\Desktop\tme-20252\bt-49495.c` and provides navigation options like Search, Run, and Terminal.